

Course Code	21IPE602J	Course Name	Semiconductor Virtual Nanofabrication	Course Category	E	Engineering Science	L	T	P	C
							3	0	2	4

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Physics and Nanotechnology		Data Book / Codes/Standards	Nil	

Course Learning Rationale (CLR):	<i>The purpose of learning this course is to:</i>
CLR-1 :	<i>Know what semiconductor process technology is and acquire special skillset for a fab engineer semiconductor industry demand</i>
CLR-2 :	<i>Prepare yourself to meet with the requirement of highly trained workforce to nurture the India Semiconductor Mission</i>
CLR-3 :	<i>Understand the basic physics of CMOS devices and various processes and tools involved in CMOS micro/nano fabrication</i>
CLR-4 :	<i>Become familiarize with the CMOS processing through virtual training using SEMulator3D</i>
CLR-5 :	<i>Study the fundamentals of Nanofabrication and Microfabrication</i>

Course Outcomes (CO):	<i>At the end of this course, learners will be able to:</i>
CO-1 :	<i>Comprehend CMOS device physics and differentiate between design, fabrication, packaging, system manufacturing and product</i>
CO-2 :	<i>Develop integral skills in semiconductor microfabrication unit processes</i>
CO-3 :	<i>Develop module-level understanding of advanced CMOS FinFET process</i>
CO-4 :	<i>Demonstrate ability to use SEMulator3D to design CMOS devices and solve integration challenges</i>
CO-5 :	<i>Familiarize with Statistical Process Control</i>

Program Outcomes (PO) (1- Low, 2 – Medium, or High-3)											
1	2	3	4	5	6	7	8	9	10	11	12
Engineering Knowledge	Problem Analysis	Design & Development	Analysis, Design, Research	Modern Tool Usage	Society & Culture	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning
3	3	-	-	-	-	-	-	-	-	-	-
3	-	3	-	-	-	-	-	-	-	-	-
-	3	-	-	3	-	-	-	-	-	-	-
-	-	-	3	3	-	-	-	-	-	-	-
-	3	-	3	-							

MODULE 1: INTRODUCTION TO SEMICONDUCTORS (12 hours) Overview of semiconductor nanofabrication – Moore's law, IRDS roadmap, semiconductor logic, memory, analog, display, and MEMS. Fundamentals of MOSFET – Device structure, working principle, and drain characteristics. Manufacturing ecosystem - difference between design, fabrication, packaging, system manufacturing and product. FEOL, BEOL. Semiconductor characterization - optical, microstructural, electrical measurements at wafer/device level. Lab 1: Install SEMulator3D and familiarize with tools Lab 2: Create a layout of NMOS, PMOS and inverter using SEMulator3D Lab 3: Imaging using electron microscopy MODULE 2: UNIT PROCESSES (12 hours) Silicon wafer – preparation, cleaning, wafer orientation. Modification - Doping (Diffusion and ion implantation), Rapid thermal processing. Deposition - CVD, PVD, ALD, electroplating. Patterning - Optical lithography, mask, multiple patterning (SADP and SAQP). Etching – dry etching, wet etching, chemical mechanical polishing (CMP), atomic layer etching. Inspection – Defect metrology, thin film thickness measurements Lab 4: Implantation & Silicidation using SEMulator3D Lab 5: Photolithography control parameters in design layout MODULE 3: PROCESS MODELING – Part I (12 hours) Isolation module – Advantages and challenges of STI verses LOCAS. Device architecture – Planar to FinFET, use of SiGe and strain. Gate module – SiO ₂ verses high-k dielectrics, polysilicon verses metal gate, gate-first verses gate last, integration challenges of replacement metal gate. Well & junction module – Basics of implantation, source-drain and well implants, other implants. Lab 6: LOCOS isolation verses Shallow Trench Isolation (STI) Lab 7: Gate-First versus Gate-Last Lab 8: Spacer and multiple patterning Lab 9: Replacement metal gate

MODULE 4: PROCESS MODELING – Part II (12 hours)

Contact module – contact resistance issues of simple metal/silicon contact, metal silicides, silicidation process, evolution from Ti to Co to Ni. Interconnect module – Need for multiple metal layers, RC delay, challenge of electromigration, evolution from Al verses Cu, Damascene and dual-damascene process, W plugs and via. Intermediate dielectric – Need for ILD, Low-k due to RC delay, SiO₂ to SiOF to SiOC to porous SiOC, material and integration challenges

Lab 11: Contact resistance measurements

Lab 12: Damascene

Lab 13: Dual-Damascene

Lab 14: PlanarFET vs FinFET inverter

MODULE 5: STATISTICAL PROCESS CONTROL (12 hours)

Introduction and importance of SPC, Basic Concepts of Statistical Distribution, Hypothesis Testing, Theory of z-test, t-test, ANOVA, Sample size estimation, Theory of DOE, Screening test, RSM, Fractional and Full-factorial tests, Paired Means.

Lab 15: Module-level understanding of CMOS 14-nm process

Learning Resources	1. S. M. Sze and M. K. Lee. “Semiconductor Devices: Physics and Technology” – John Wiley & Sons Inc., 3rd edition, 2012. 2. Donald A. Neamen, “Semiconductor Physics and Devices – Basic Principles” – McGraw Hill, 4th Edition, 2012 3. Yoshio Nishi and Rober Doering (Eds.), “Handbook of Semiconductor Manufacturing Technology”- CRC Press, 2nd Edition, 2007.	4. Marc J.Madou. “Fundamentals of Microfabrication and Nanotechnology – Volume II” – CRC Press, 3rd Edition, 2011. 5. Gary S. May and Costas J. Spanos “Fundamentals of Semiconductor Manufacturing and Process Control” IEEE A JOHNWILEY&S ONS, INC., Publication. 6. Badih El-Karch “Fundamentals of Semiconductor Processing Technology” Springer Nature. 7. https://onlinecourses.nptel.ac.in/noc20_bt37/preview. 8. https://onlinecourses.nptel.ac.in/noc20_bt17/preview.
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	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)					
		Formative CLA-I Average of unit test (20%)		Life Long* Learning CLA-II- Practice (40%)		Summative Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20	-	-	-	-	-
Level 2	Understand	20	-	-	20	-	20
Level 3	Apply	30	-	-	30	-	30
Level 4	Analyze	30	-	-	30	-	30
Level 5	Evaluate	-	-	-	20	-	20
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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